

Chapter 6

NMR data acquisition and inversion

Processing of NMR data involves detailed knowledge of the data acquisition procedure, how different pulsing sequences are averaged, and the proper phase rotation to obtain the echo train signal. There are also various schemes for data inversion to obtain the T_2 distributions which deserve a careful review.

In the following, we shall describe the general procedure of acquiring NMR data for T_2 CPMG type measurement in the laboratory, and follow with a discussion on various data inversion methods. Different NMR logging tools have slightly different pulsing sequences. As a result, the way the signals are averaged may affect the data storage format. The readers can refer to vendor's manuals for details. However, the basic general principles of CPMG data acquisition are the same.

6.1 Data acquisition

In addition to the detection of the FID and the CPMG spin echo signals, we need to digitize, store, and process these signals. For a typical CPMG spin echo train, there are frequently hundreds or even thousands of echoes. It is not possible, and not necessary either, to digitize the entire echo profile, only the peaks of the spin echo signal need to be measured and stored. There are two ways to record the signal: the single channel and the quadrature detection methods.

6.1.1 Signal detection

In the single detection method, the rf signal from the sample is fed into a single phase-sensitive detector for determining the frequency difference between the sample signal and the carrier frequency of the rf pulse (i.e., the mean Larmor frequency which we set for the measurements). Since a single phase-sensitive detector can determine only the magnitude and not the sign of this difference, we won't be able to determine whether the offset frequency is positive or negative. Hence, a single spectral feature, after Fourier transform, can be detected at two different frequencies symmetrically located about the carrier frequency ν_0 . The process of reflecting the feature at, say, $\nu_0 + \delta\nu$ to $\nu_0 - \delta\nu$, is called folding.

To avoid confusion, the offset of the carrier frequency must be chosen at one end of the spectral region. Signals which occur outside the range of the measurement become "folded" and appear as mirror signals, usually of indeterminate phase.

Because of its inability to determine the phase differences, single detection is not used in NMR logging. Rather, quadrature detection is the method normally used for signal detection.

Using quadrature detection, the incoming sample signal is fed to two identical phase-sensitive detectors whose reference signals differ by exactly 90° . The resultant audio signals are amplified, passed through identical low-pass filters, digitized by a multiplexed analog to digital converter and stored in separate data memory blocks. Quadrature Fourier transformation produces a real and an imaginary spectrum in a manner analogous to single detection with the exception that now plus and minus frequencies relative to the rf carrier frequency can be distinguished.

Suppose the incoming sample signal, after mixing with the local oscillator frequency ω_c , has the form $\cos[(\omega_0 - \omega_c + \Delta\omega_B)t - \phi]$, where ω_0 is the mean Larmor frequency, ω_c is the carrier frequency, and $\Delta\omega_B$ is the bandwidth for all resonant spins. When $\omega_0 = \omega_c$, the mixed signal has the following form:

$$R(t) = \sum_i A_i(t) \cos(\omega_i t - \phi) \quad (6.1)$$

where the summation is over all frequencies ω_i within the resonance bandwidth from $-\Delta\omega_B/2$ to $+\Delta\omega_B/2$, and $A_i(t)$ is the amplitude for ω_i . When this is fed to the two phase-sensitive detectors, there would be a cos (real)

channel as Eq. (6.1), and a sin (imaginary) channel as

$$I(t) = \sum_i A_i(t) \sin(\omega_i t - \phi) \quad (6.2)$$

In the case of a FID, $A_i(t)$ will be an exponential decay function. The rapid decay of the FID is the result of the summation of various frequencies within the resonance band. In the case of CPMG measurements, the sample signal will have the appearance as shown in Figure 6.1. Notice that (a) and (b) of Figure 6.1, representing the output of the two phase-sensitive detectors, have exactly the same envelop $\sum_i A_i(t)$. This is a case where the phase angle ϕ is exactly zero. Hence, if we digitize the output curve at a time interval of 2τ , i.e., where the peak of the spin echo is, Figure 6.1(a) will be $\sum_i A_i(t) \cos 0^\circ$, i.e., tracing out the profile of $\sum_i A_i(t)$, and Figure 6.1(b) will be $\sum_i A_i(t) \sin 0^\circ$, tracing out the dc base line.

In the T_2 CPMG data storing process, only the signal within a very small time window at the peak of the spin echo is digitized for storage. Hence, Figures 6.1(a) and (b) will be further simplified to Figures 6.2(a) and (b). Figure 6.2(a) shows the T_2 echo train signal without any noise, and it is on this one that we perform our NMR inversion computation to obtain the T_2 amplitude distribution. Figure 6.2(b) simply shows the dc base line. Of course, in real log data, both of them will be masked by noise fluctuations.

If the phase angle ϕ is not zero, then the two channels will have the appearance as shown in Figure 6.3, and after sampling the signal every 2τ , they become Figure 6.4, which shows the two channels of data normally acquired during the NMR logging run.

The T_2 CPMG measurements of the logging runs are always collected in pairs, known as Phase-Alternated Pairs (PAPs), in order to cancel the ringing and baseline offset. The first echo train is acquired with the normal settings, whereas in the second set the phase of the transmitter pulse is changed 180° to give spin echoes of negative amplitude. The two sets of CPMGs are referred to as the plus phase and minus phase. These pairs are combined to give PAPs which eliminate the effects of ringing and baseline offset. In the laboratory where measurement time is not an issue, more refined measurements are possible and signals are almost always collected in four quadrants, instead of just two, to completely eliminate phase offset errors.

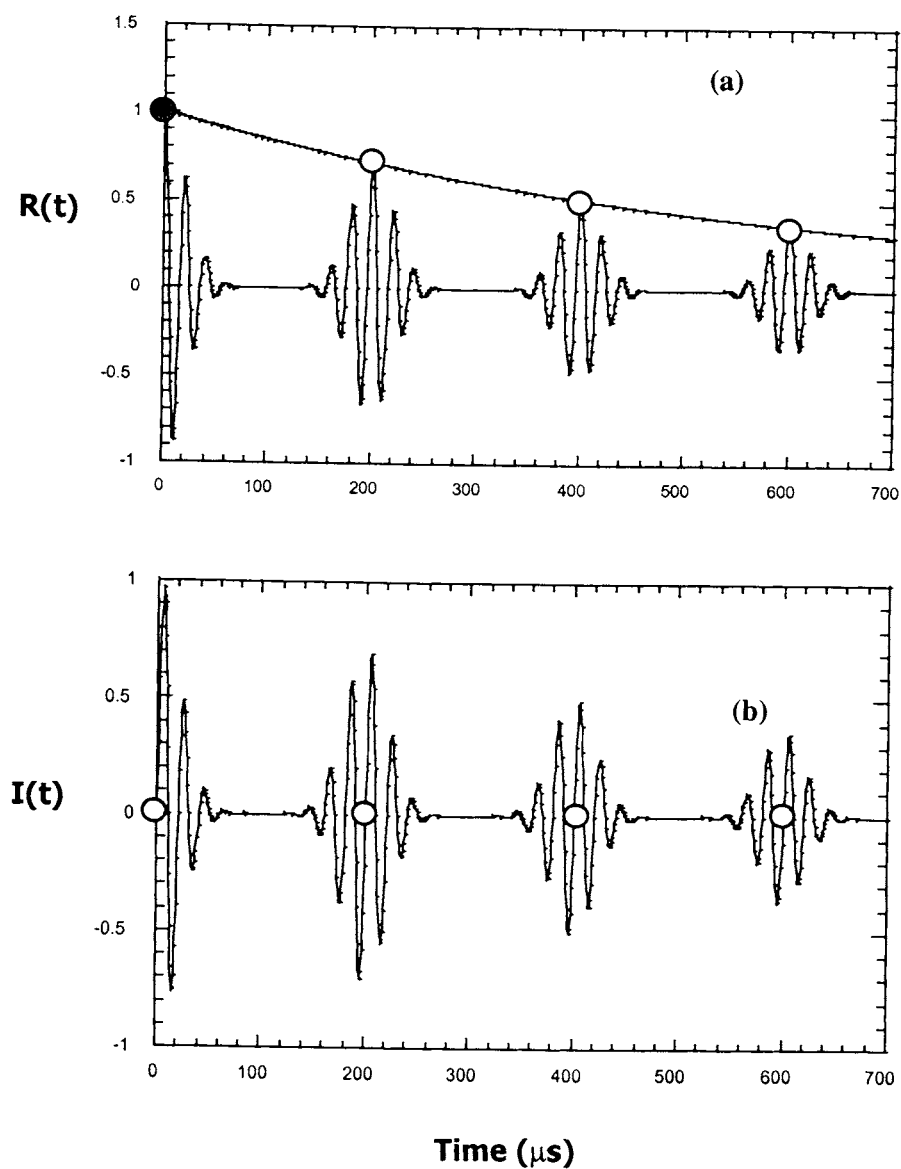


Figure 6.1: The two quadrature channels (a) $\cos$ (real) and (b) $\sin$ (imaginary) when the phase angle ϕ is zero

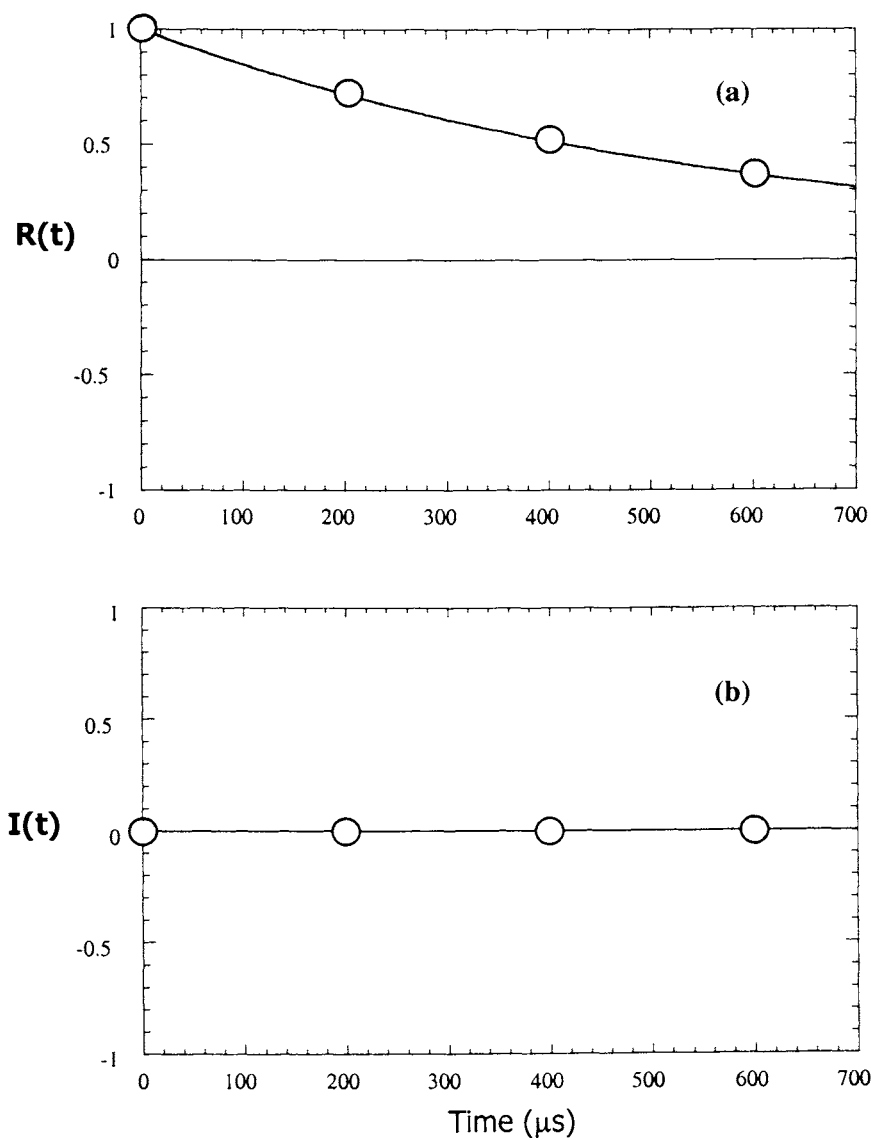


Figure 6.2: The two quadrature channels of Figure 6.1 after collecting the signal at the peak of the spin echo.

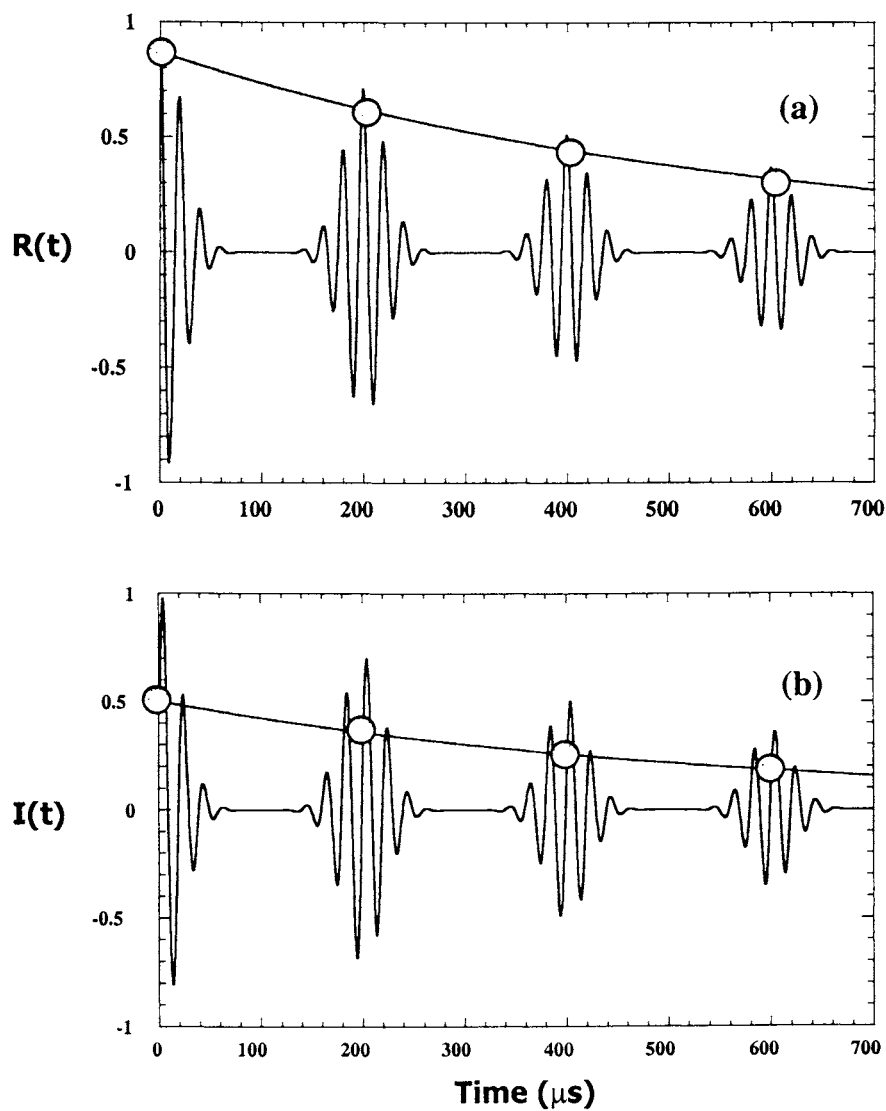


Figure 6.3: The two quadrature channels (a) $\cos$ (real) and (b) $\sin$ (imaginary) when the phase angle ϕ is not zero

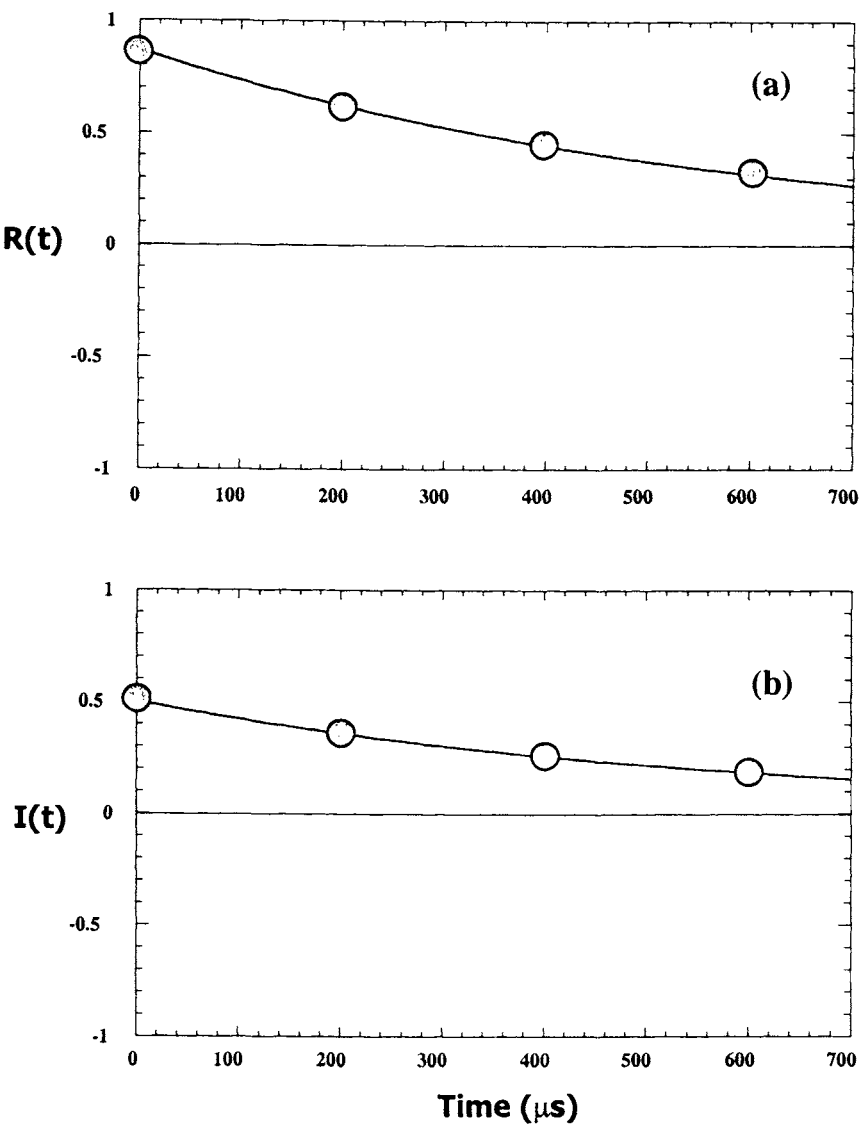


Figure 6.4: The two quadrature channels of Figure 6.3 after collecting the signal at the peak of the spin echo.

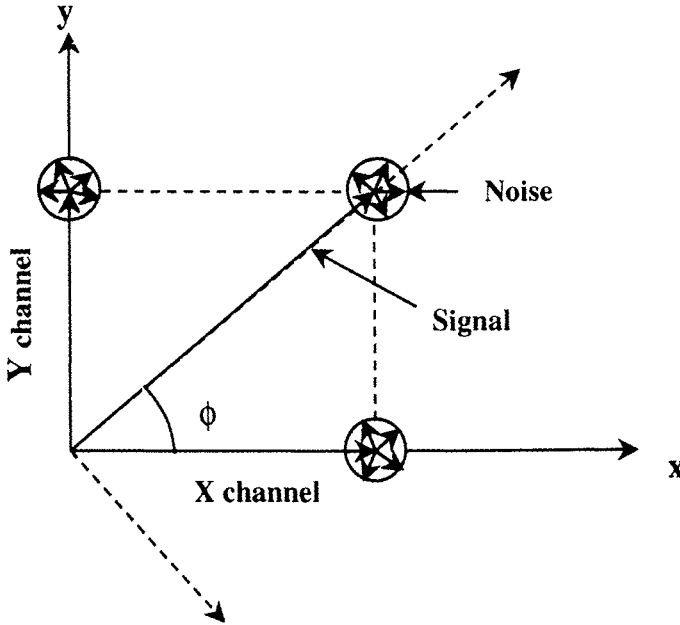


Figure 6.5: The echo signal magnitude and noise in polar coordinates and their relationship to X and Y channels.

6.1.2 Phase rotation

The two channels of recorded data are typically referred to as X and Y channels. To compute the phase angle (see for example, Freedman [88]), we do the following,

$$X_j = S_j \cos \phi + \varepsilon_j^X \quad (6.3)$$

$$Y_j = S_j \sin \phi + \varepsilon_j^Y \quad (6.4)$$

where X_j and Y_j are the amplitudes of the j -th echo in the X and Y channels, respectively, S_j the true magnitude of the echo, ϕ the phase angle, and ε_j^X and ε_j^Y the noise for the j -th echo in X and Y channels, respectively. Figure 6.5 shows their relationship in polar coordinates. If we sum up all echoes (assuming n echoes) in the echo train, we get,

$$\sum_{j=1}^n X_j = \left(\sum_{j=1}^n S_j \right) \cos \phi + \left(\sum_{j=1}^n \varepsilon_j^X \right) \quad (6.5)$$

$$\sum_{j=1}^n Y_j = \left(\sum_{j=1}^n S_j \right) \sin \phi + \left(\sum_{j=1}^n \varepsilon_j^Y \right) \quad (6.6)$$

Because the noise is random with equally probable positive and negative values, the second term in Eqs. (6.5) and (6.6) eventually approaches zero when the averaged noise is computed. Taking the ratio of (6.6) to (6.5), we have,

$$\phi = \tan^{-1} \left(\frac{\sum_{j=1}^n Y_j}{\sum_{j=1}^n X_j} \right) \quad (6.7)$$

Once the phase angle is known, the coordinates can be rotated using the following equations to obtain the signal channel and the noise channel, as shown in Figure 6.6

$$X_j \cos \phi + Y_j \sin \phi = S_j + (\varepsilon_j^X \cos \phi + \varepsilon_j^Y \sin \phi) \quad (6.8)$$

$$-X_j \sin \phi + Y_j \cos \phi = (-\varepsilon_j^X \sin \phi + \varepsilon_j^Y \cos \phi) \quad (6.9)$$

where (6.8) represents rotation to the signal channel, (6.9) rotation to the noise channel, and the quantities within the parentheses in (6.8) and (6.9) are noise.

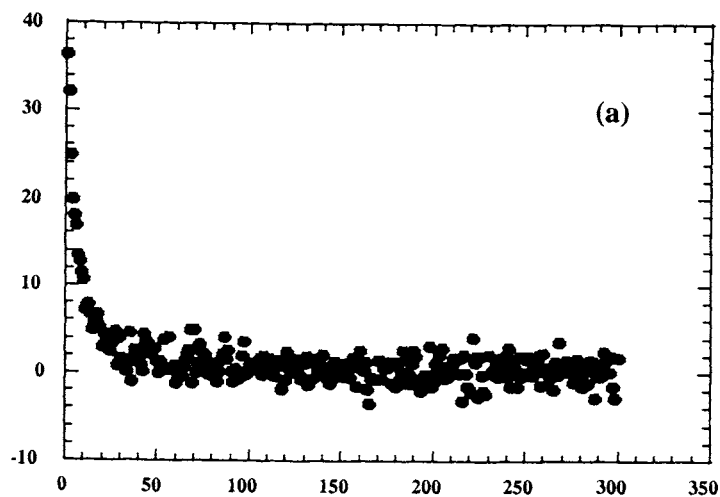
In most cases, the summation in Eqs. (6.5) and (6.6) is performed from the first through the last echoes to have good statistics in canceling the noise. This is especially true for large signals in which $\sum X_j$ (or $\sum Y_j$) is dominated by $\sum S_j \cos \phi$ (or $\sum S_j \sin \phi$). However, when we encounter shaly zones where porosities are small, the signal diminishes quickly after a few spin echoes and the rest of the echo train is just noise. The summation over noise, $\sum \varepsilon_j^X$ (or $\sum \varepsilon_j^Y$), actually has a standard deviation which increases as $\sqrt{n}$, even though the averaged noise, i.e., $\langle \varepsilon^X \rangle = (\sum \varepsilon_j^X)/n$ decreases as $1/\sqrt{n}$. Hence, in these situations, the computation of phase angle may not be accurate if all echoes are taken. The solution for this is to perform the summation up to the point where the echoes approach the baseline, then stop the summation and ignore the rest of the echoes which are essentially noise.

Sometimes, instead of X_j and Y_j , the echo magnitude $A_j = \sqrt{X_j^2 + Y_j^2}$ and the echo phase $\psi_j = \tan^{-1}(Y_j/X_j)$ for each individual echo are provided. Then, one has to compute X_j and Y_j from A_j and ψ_j and then perform the proper rotation to get the signal channel.

6.2 Data inversion

We have discussed previously in Chapter 4 that there are non-linear inversion schemes which use a two or three exponential model, and linear inversion

ECHO AMPLITUDES



NOISE AMPLITUDES

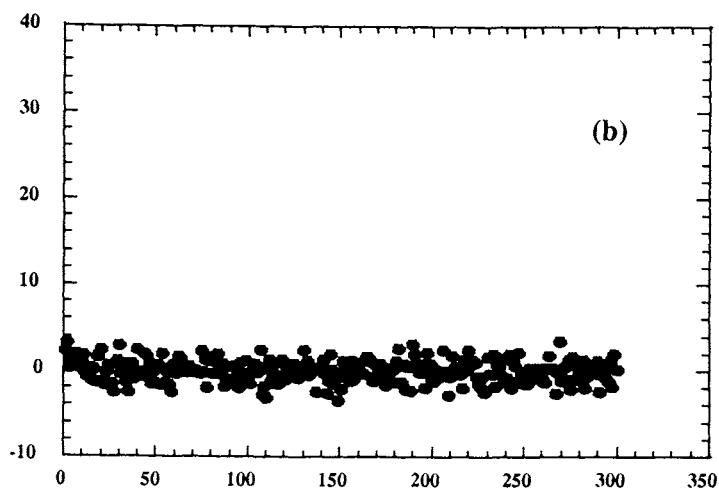


Figure 6.6: The signal and noise channels of a typical log measurement after the phase rotation.

schemes which assume a distribution of known T_2 relaxation times equally spaced on a logarithmic scale and solve for amplitudes. The CPMG T_2 echo trains obtained from NMR logging commonly use the linear inversion scheme to obtain T_2 amplitude distributions. We shall restrict our discussion here to the linear inversion scheme.

The NMR logging tools typically acquire T_2 CPMG spin-echo trains of several hundreds to a few thousands of echoes. The inversion process involved is to find a set of T_2 amplitudes, f_j , from a set of measurements of echoes, g_i :

$$g_i = M(t_i)/M_0 = \sum_{j=1}^m a_j f_j e^{-t_i/T_j} + \epsilon_i, \quad i = 1, \dots, n \quad (6.10)$$

by minimizing the error ϵ_i . Here we assume that there are n echoes each with an amplitude g_i measured at t_i and m known (i.e., assumed) relaxation times T_j preselected to be equally spaced on a logarithmic scale. The echo amplitude g_i is given by the magnetization $M(t_i)$ of the system measured at t_i normalized by the magnetization at $t = 0$, M_0 . In NMR logging data, the echo amplitudes are typically in porosity units (PU).

We have assumed that f_j is the partial porosity for the pores which have a T_2 relaxation time of value T_j associated with them. Note that we have a factor, a_j , besides f_j in Eq. (6.10). This factor a_j is the polarization factor, and is given by

$$a_j = 1 - e^{-RD/T_{1j}} \quad (6.11)$$

where RD is the repeat delay time or wait time. It is the time between successive exciting pulses for the CPMG echo sequences. T_{1j} is the T_1 relaxation time associated with these pores of size j .

Normally, we use a long enough wait time, typically three times the T_{1j} relaxation time to allow all the spins in the system to fully recover to their equilibrium state. In this situation, the factor a_j is one. Sometimes, we use a very short wait time purposely to measure only the short relaxation time components while suppressing the long relaxation time components. In this case, a_j will still be close to one for short time components, but can be suppressed to a fraction of one or near zero for long time components. When solving for f_j , this polarization factor is lumped into the factor e^{-t_i/T_j} by assuming a known T_{1j}/T_{2j} ratio (note that, in Eq. (6.10), T_{2j} is simplified as T_j). Thus, the polarization factor is given by

$$a_j = 1 - e^{-RD/rT_j} \quad (6.12)$$

where r is the T_{1j}/T_{2j} ratio. In the following discussion, we shall assume a_j is equal to 1.

Since all e^{-t_i/T_j} 's are known and we are solving for f_j , it is a linear inversion problem. A least squares fit is used to minimize the following sum:

$$\min \left\{ \phi(f) = \sum_{i=1}^n \frac{1}{\sigma_i^2} \left(\sum_{j=1}^m f_j e^{-t_i/T_j} - g_i \right)^2 \right\}, \quad (6.13)$$

where σ_i is the measurement error of the i -th echo, g_i .

This problem is well known to be *ill-posed*. Small fluctuations in the measured data can lead to drastically different sets of f_j 's which all give reasonably good fits. It requires a procedure, known as *regularization*, to obtain a reasonable solution. To have a clear understanding of these terms, *ill-posed* and *regularization*, we shall discuss this further in the following section.

6.2.1 Ill posed problem

The subject of *ill-posed*, or *incorrectly-posed* problems has been studied extensively in the past by Tikhonov [211], Turchin [218], Phillips [175], Twomay [219], Wahba [221], and Butler et al. [48]. To define this in precise mathematical terms, we shall follow Tikhonov [211].

The problem $\mathcal{R}$ of the determination of a function $f(s) \in \mathcal{F}$ by means of a given function $g(t) \in \mathcal{G}$, $f(s) = \mathcal{R}[s, g(t)]$ (where $\mathcal{F}$ and $\mathcal{G}$ are function spaces), is said to be *correctly-posed* if:

1. For every function $g(t) \in \mathcal{G}$, there exists a solution $f(s)$ of the problem.
2. The solution $f(s)$ is unique for a given $g(t)$.
3. The dependence of the solution $f(s)$ on $g(t)$ is continuous in the metrics of $\mathcal{F}$ and $\mathcal{G}$

The problem $\mathcal{R}$ is said to be *incorrectly-posed* or *ill-posed*, if the above conditions cannot be satisfied. For example, in a *correctly-posed* problem, for a given function $\bar{g}(t)$, there corresponds a unique solution $\bar{f}(s) = \mathcal{R}[s, \bar{g}(t)]$ for the given problem $\mathcal{R}$. Moreover, even if we know $\bar{g}(t)$ only approximately, we can still determine $\bar{f}(s)$ reliably in the following sense: For any $\epsilon > 0$, no matter how small, there exists a $\delta > 0$ such that if $\|g(t) - \bar{g}(t)\| < \delta$, then the exact unique solution $f(s) = \mathcal{R}[s, g(t)]$ satisfies $\|f(s) - \bar{f}(s)\| < \epsilon$. In an

ill-posed problem, by contrast, it is not possible, in general, to control the accuracy of $\tilde{f}(s)$ in this fashion. In some cases, even the existence and/or uniqueness of $\tilde{f}(s)$ is not ensured.

Many important practical problems are ill-posed. For instance, Fredholm's equation of the first kind is one of the most important examples. Eq. (6.10) is an example of a Fredholm equation of the first kind in discrete form. We can recast that in the following integral form with continuous variables and functions:

$$\hat{K}f \equiv \int_a^b K(t, s)f(s)ds = g(t), \quad c \leq t \leq d \quad (6.14)$$

where e^{-t_i/T_j} is replaced by $K(t, s)$, f_j by $f(s)$, and g_i by $g(t)$, and s stands for the relaxation time T . The problem becomes to solve for $f(s)$, given $g(t)$ and the functional form of $K(t, s)$.

In this general form, $K(t, s)$ is called the kernel of the equation. It can be any arbitrary function, which determines the operator $\hat{K}$ of the problem and converts the unknown function $f(s)$ into some other function $g(t)$, which is accessible to physical measurements. The problem itself, in terms of the definitions above, is denoted as $\mathcal{R}$.

Frequently, the solution for Eq. (6.14) may not exist at all, or may not exist for every function $g(t)$ on the right hand side. For instance, if

$$K(t, s) = t + s \quad (6.15)$$

the right hand member $g(t)$ must be a linear function of t . If $g(t) = g_0 + g_1 t$, then any function $f(s)$ that satisfies the following conditions,

$$\int_a^b f(s)ds = g_1, \quad \int_a^b s f(s)ds = g_0, \quad (6.16)$$

is a solution of Eq. (6.14) with kernel (6.15). It is clear that the solution for $f(s)$ is not unique, and the problem is consequently *ill-posed*.

Sometimes, in problems arising from physics, we are fairly sure of the existence and uniqueness of the function $f(s)$ (e.g., the inverse problem of potential theory). However, even in this case, Eq. (6.14) possesses a peculiar property which makes the problem an *ill-posed* one. This peculiar property is the "smoothing" action of the kernel. We shall illustrate this with an example.

Let the kernel $K(t, s)$ be a continuous function of its second argument s . Let us assume that there exists a solution $f_1(s)$ for a given $g_1(t)$. Then the

function $f_2(s) = f_1(s) + C \sin 2\pi ns/(b-a)$, where C is a large number and n is a sufficiently large integer, can also be made an approximate solution of the problem. It is quite clear that for any arbitrarily large C , we can find a large enough n such that the absolute value of the difference of the corresponding right hand members $g_1 = \hat{K}f_1$ and $g_2 = \hat{K}f_2$ can be made arbitrarily small. Thus the kernel in question has “smoothed out” a very intense, but very high frequency component to an extremely small level. If the right hand member $g(t)$ could be measured exactly, then the uniqueness of the solution $f(s)$ would have been guaranteed. However, measurement error always exists in obtaining $g(t)$. Thus wildly different $f(s)$, such as $f_1(s)$ and $f_2(s)$ always exist, and in fact, there are infinitely many such solutions. The problem has become *ill-posed*.

In practical physical problems, such as the NMR data inversion discussed here, the numerical solution of Eq. (6.14) always reduces it in one way or another to a system of linear algebraic equations

$$(\hat{K}f)_i \equiv \sum_{j=1}^m K_{ij}f_j = g_i, \quad i = 1, 2, \dots, n \quad (6.17)$$

where K_{ij} , f_j , and g_i are discrete values of their respective continuous functions, and g_i 's are accessible for measurements while f_j are to be determined.

Thus the uniqueness of the solution depends strongly on the precise measurement of g . Any error in g can lead to drastically different $f(s)$. It has been shown (Fadeev [81], Lanczos [131]) that the stability of the solution for such an inversion problem is closely related to the set of eigenvalues μ of the matrix K^TK , where K and K^T are the matrix forms of the operator $\hat{K}$ and the adjoint operator $\hat{K}^\dagger$, respectively. The stability decreases as the ratio $\mu_{\max}/\mu_{\min}$ increases. Consequently, the demand for measurement precision and accuracy for $g(t)$ becomes greater and greater. The measurement error propagates to all components of the solution $f(s)$ through a factor proportional to μ^{-1} . Thus as the μ we include in the inversion gets smaller, the solution becomes more unstable. We shall discuss this further in the section on singular value decomposition.

Faced with such a situation, it is really hopeless mathematically, unless we can inject new information into the process and place some kind of physical constraints on the solution $f(s)$ based on a priori knowledge about the nature of $f(s)$. One such constraint is the belief that the function $f(s)$ should be reasonably smooth. To incorporate this into the inversion process, the general

approach is to add a penalty function in Eq. (6.13) to smooth or *regularize* the solution. This *regularization* process can be defined more precisely as follows.

We shall call a process $\mathcal{R}_\delta$ a regularizing process (or algorithm) if :

1. $\mathcal{R}_\delta[s, \bar{g}(t)]$ is defined for every $\bar{g} \in \mathcal{G}$ and $\delta > 0$.
2. If for $\bar{g}(t)$ there exists a $\bar{f}(s) = \mathcal{R}[s, \bar{g}(t)]$, then for every positive ϵ there exists a $\delta(\epsilon, \bar{f})$ such that if $\|\tilde{g}(t) - \bar{g}(t)\|_{\mathcal{G}} < \delta$, then $\|\tilde{f}_\delta(s) - \bar{f}(s)\|_{\mathcal{F}} \leq \epsilon$, where $\tilde{f}_\delta = \mathcal{R}_\delta[s, \tilde{g}]$.

We shall say that a problem $f(s) = \mathcal{R}[s, g(t)]$ is a *regularizable* problem if it has at least one regularizing process.

Here we introduce three types of *regularization* depending on the types of penalty function, namely: the “norm” (or the “amplitude”) smoothing,

$$\min \left\{ \phi(f) = \frac{1}{2} \sum_{i=1}^n \left(\sum_{j=1}^m f_j k_{ij} - g_i \right)^2 + \frac{\alpha}{2} \sum_{j=1}^m f_j^2 \right\} \quad (6.18)$$

the “slope” smoothing,

$$\min \left\{ \phi(f) = \frac{1}{2} \sum_{i=1}^n \left(\sum_{j=1}^m f_j k_{ij} - g_i \right)^2 + \frac{\alpha}{2} \sum_{j=1}^m (f_{j+1} - f_j)^2 \right\} \quad (6.19)$$

and the “curvature” smoothing,

$$\min \left\{ \phi(f) = \frac{1}{2} \sum_{i=1}^n \left(\sum_{j=1}^m f_j k_{ij} - g_i \right)^2 + \frac{\alpha}{2} \sum_{j=1}^m (f_{j+1} + f_{j-1} - 2f_j)^2 \right\} \quad (6.20)$$

where $k_{ij} = e^{-t_i/T_j}$, and α , often referred to as a “smoothing” or a “regularization” parameter, is a constant optimally chosen to be commensurate with the measurement error. Recent work by Borgia, et al [34] extends the treatment on the “curvature” smoothing to provide a uniform penalty along the relaxation time axis by allowing the constant α to be a variable.

Other regularizing processes such as the basis function method and singular value decomposition method do not specifically have a penalty function. However, there are also regularizing actions involved in the selection of basis functions or singular values. Although there are many regularization processes for the NMR CPMG data inversion problem, the characteristic shape

of the T_2 distributions obtained by various methods are by and large similar, and the total sums of the amplitudes are generally in agreement. For a specific regularization process, the measurement error δ determines the uncertainty of the solution ϵ . There may be slight difference in the shape of the T_2 distributions (i.e. $f(s)$). This is because different regularization processes impose different constraints on the infinitely many possible solutions $f(s)$.

6.2.2 Norm smoothing

There are multiple methods for implementing “norm” smoothing. One commonly used procedure for minimizing Eq. (6.18) is the Levenberg-Marquardt method [180] with the non-negative constraint for all f_j ’s in the iterative process. The general concern, however, is how to choose an optimal value for the smoothing parameter α to be commensurate with the measurement error.

A frequently used criterion is to minimize the difference between the vector $f^{(\alpha)}$ (the α -dependent solution from Eq. (6.18)) and $f^{(0)}$ (the true solution), and somehow relate that to the known measurement error of the echo amplitudes. With the input of the known measurement error, the optimal value for α is then determined.

Here, we shall present a method proposed by Butler, Reeds, and Dawson (BRD) [48]. The following discussion is based on a simplified description by Bergman [15] of the original Butler-Reeds-Dawson paper.

The problem is to minimize the following quantity

$$\phi(f_x) = \phi(f_1 \dots f_m) \equiv \frac{1}{2} \sum_{i=1}^n \left(\sum_{x=1}^m k_{ix} f_x - g_i \right)^2 + \frac{\alpha}{2} \sum_{x=1}^m f_x^2. \quad (6.21)$$

by finding a set of f_x which satisfy the positivity constraint

$$f_x \geq 0 \text{ for all } x.$$

Eq. (6.21) is minimized when the following set of $n+m$ coupled equations are satisfied by the $n+m$ unknowns $f_x^{(\alpha)}$, c_i

$$\sum_{y=1}^m k_{iy} f_y^{(\alpha)} - g_i + \alpha c_i = 0, \quad i = 1 \dots n \quad (6.22)$$

$$f_x^{(\alpha)} = \max \left(0, \sum_{i=1}^n c_i k_{ix} \right), \quad x = 1 \dots m. \quad (6.23)$$

To show this we note that the constrained minimum of $\phi(f_x)$ can be found by considering its m -dimensional gradient, given by

$$(\nabla\phi)_x \equiv \frac{\partial\phi}{\partial f_x} = \sum_{i=1}^n k_{ix} \left(\sum_{y=1}^m k_{iy} f_y - g_i \right) + \alpha f_x. \quad (6.24)$$

At the calculus minimum, all components of $\nabla\phi$ must vanish. However, if some of the f_x happen to be negative, then the *constrained* minimum will have some of these components (not necessarily the same components that are negative at the calculus minimum!) set to their extreme allowed nonnegative values, namely 0. At such a *non-calculus* minimum, the components of $\nabla\phi$ whose corresponding f_x were made 0 will be *positive*, while the other components of $\nabla\phi$ whose corresponding f_x were positive, vanish. Hence at the constrained minimum

$$\begin{aligned} (\nabla\phi)_x &= 0 \quad \text{for } f_x > 0 \\ (\nabla\phi)_x &\geq 0 \quad \text{for } f_x = 0. \end{aligned} \quad (6.25)$$

The gradient of ϕ can be calculated for $f_x = f_x^{(\alpha)}$, which are solutions of (6.22) and (6.23): We multiply (6.22) by k_{ix} , sum over i , and obtain

$$0 = \sum_{i=1}^n k_{ix} \left(\sum_{y=1}^m k_{iy} f_y^{(\alpha)} - g_i \right) + \alpha \sum_{i=1}^n c_i k_{ix}. \quad (6.26)$$

If $f_x > 0$, then the last term in this equation is equal to αf_x , and we get $(\nabla\phi)_x = 0$. If $f_x = 0$, then the last term in this equation must be negative, and (6.23)–(6.26) imply that $(\nabla\phi)_x \geq 0$. This proves that the solution of (6.22) and (6.23) minimizes (6.21) and satisfies the positivity constraint (6.25). To solve the coupled equations (6.22) and (6.23), we note that they can be modified to the following set of n equations for the c_i

$$\sum_{j=1}^n M_{ij} c_j + \alpha c_i = g_i, \quad (6.27)$$

where the matrix $\hat{M}$ is defined by

$$M_{ij} \equiv \sum'_x k_{ix} k_{jx}. \quad (6.28)$$

and $\sum'_x$ stands for a restricted sum over those values of the index x for which $\sum_{i=1}^n c_i k_{ix} > 0$. Because of this restriction, the matrix $\hat{M}$ depends upon the c_i , and (6.27) is *not* a set of linear equations.

Eqs. (6.27) can be solved in a number of ways. One way is to note that the solution also minimizes the following quantity

$$\psi \equiv \frac{1}{2}c^\dagger(\hat{M} + \alpha\hat{I})c - c \cdot g, \quad (6.29)$$

and proceed to find a vector c at which ψ is a minimum. Another method for solving (6.27) is to start with a full matrix $M_{ij} = \sum_x k_{ix}k_{jx}$ without the restriction on the summation, solve for c , then update M_{ij} according to (6.28), and iterate to the final correct solution.

The problem of optimizing the regularization parameter α is handled by attempting to minimize the difference between the vector $f^{(\alpha)}$ and $f^{(0)}$, i.e.

$$(f^{(\alpha)} - f^{(0)})^\dagger \cdot (f^{(\alpha)} - f^{(0)}) = \sum_x (f_x^{(\alpha)} - f_x^{(0)})^2 \quad (6.30)$$

where $f^{(0)}$ is the “true” f being sought. Since $f^{(0)}$ is independent of α , this is equivalent to minimizing

$$\sum_x (f_x^{(\alpha)} - f_x^{(0)})^2 - \sum_x (f_x^{(0)})^2 = \sum_x (f_x^{(\alpha)})^2 - 2 \sum_x f_x^{(\alpha)} f_x^{(0)}. \quad (6.31)$$

We then develop the last, unknown term as follows

$$\sum_x f_x^{(\alpha)} f_x^{(0)} = \sum_i \sum'_x c_i k_{ix} f_x^{(0)} \geq \sum_i c_i g_i^{(0)}, \quad (6.32)$$

where $g_i^{(0)}$ denotes the ideal spin-echo amplitudes with no noise. If we denote the noise by ϵ_i , we finally get the following upper bound for (6.31), using (6.23) and (6.28),

$$c^\dagger \hat{M} c - 2c \cdot g^{(0)} = c^\dagger \hat{M} c - 2c \cdot g + 2c \cdot \epsilon, \quad (6.33)$$

where $g_i = g_i^{(0)} + \epsilon_i$.

This quantity, which depends on α (through $\hat{M}$ and c_i) and on ϵ_i , must be minimized by varying α over positive values. The lack of knowledge about $f_x^{(0)}$ has been translated into a lack of knowledge about ϵ_i , which is more amenable to estimations.

Assuming that the ϵ_i have the character of random noise, i.e., that they are independently distributed random variables with zero mean and identical variances σ^2 , a reasonably conservative estimate of the “maximum likely value” of $c \cdot \epsilon$ is

$$\sqrt{n}(c \cdot c)^{1/2}\sigma.$$

When this is substituted in (6.33), the resulting expression

$$H(\alpha) \equiv c^\dagger \hat{M} c - 2c \cdot g + 2\sqrt{n}\sigma(c \cdot c)^{1/2} \quad (6.34)$$

is entirely determined by the solution of (6.27). The function $H(\alpha)$ has a derivative given by

$$\frac{1}{2}H'(\alpha) = c^\dagger(\hat{M} + \alpha\hat{I})^{-1}c \left(\alpha - \frac{\sqrt{n}\sigma}{(c \cdot c)^{1/2}} \right),$$

the sign of which is determined by whether

$$\xi(\alpha) \equiv \alpha^2(c \cdot c) = \alpha^2 g^\dagger(\hat{M} + \alpha\hat{I})^{-2}g = (g - \hat{M}c)^\dagger \cdot (g - \hat{M}c) > 0 \quad (6.35)$$

is greater or smaller than the constant value $n\sigma^2$. Note that $\alpha c = g - \hat{M}c$, where $\hat{M}c = \sum_x k_{ix} f_x$ represents all vectors expressible as such a summation with some values of f_x .

Using the derivative of $\xi(\alpha)$, given by

$$\xi'(\alpha) = \alpha c \cdot c - \alpha^2 c^\dagger(\hat{M} + \alpha\hat{I})^{-1}c = \alpha c^\dagger(\hat{M} + \alpha\hat{I})^{-1}\hat{M}c > 0, \quad (6.36)$$

it is easy to construct a simple procedure for finding where $\xi(\alpha) = n\sigma^2$, which is a unique calculus minimum of $H(\alpha)$ if it exists, and provides the optimum value of α , α_{opt} .

In general, since $\xi(\alpha)$ is a monotonically increasing function, the existence of such a finite, positive value for α is assured if

$$\xi(0) < n\sigma^2 < \xi(\infty). \quad (6.37)$$

From Eq. (6.35) we get

$$\xi(\infty) = g \cdot g, \quad \xi(0) = (g - g^*) \cdot (g - g^*),$$

where g^* is the closest point to g in the space of all n -dimensional vectors expressible as $\sum_x k_{ix} f_x$ for some values of f_x . In the usual case, when

$$(g - g^*) \cdot (g - g^*) < n\sigma^2 < g \cdot g,$$

Eq. (6.37) is satisfied and a unique, finite value for α_{opt} exists and can be found using the above mentioned procedure (i.e., a procedure which finds an α_{opt} such that $\xi(\alpha_{opt}) = n\sigma^2$). If the signal-to-noise ratio is too small, so

that $n\sigma^2 > g \cdot g$, we get $\alpha_{opt} = \infty$ and a meaningless, flat distribution. If $(g - g^*) \cdot (g - g^*) > n\sigma^2$, our choice of relaxation time spectrum was bad since it is incapable of reproducing the measured spin-echo amplitudes to a good approximation, and we get $\alpha_{opt} = 0$. If this occurs, it means that we should either improve our measurement accuracy or use a different model.

There are at least two ways of implementing the BRD method. The first one suggested by Bergman [15] uses the fact that the elements of $\hat{M}$ are piecewise constant when the c_i 's change, and it works as follows:

- (1) For a fixed α , we find a solution c which satisfies

$$(M_{ij} + \alpha\delta_{ij})c_j = g_i \quad (6.38)$$

where $M_{ij} = \sum_{x=1}^m k_{ix}k_{jx}$.

- (2) The f_x is then given by

$$f_x = \max(0, \sum_{i=1}^n c_i k_{ix}). \quad (6.39)$$

(3) We then update the matrix element $M_{ij} = \sum'_x k_{ix}k_{jx}$. This time it is summed over only those x 's which give positive values for $\sum_{i=1}^n c_i k_{ix}$. The new M_{ij} is used in (1) to solve for a new c .

(4) Repeat the process, until c stops changing. The final value for f_x is given by (2).

This procedure usually works quite well. However, it is possible that direct iterations of Eq. (6.38) to solve for c may be impossible. In that case, the second procedure which was suggested by Butler et al. should be used, i.e., to search for the minimum of the convex function

$$\psi = \frac{1}{2}c^\dagger(\hat{M} + \alpha\hat{I})c - c \cdot g \quad (6.40)$$

This can be accomplished by evaluating

$$\psi' \equiv \frac{\partial\psi}{\partial c_i} = \sum_j (\hat{M} + \alpha\hat{I})_{ij}c_j - g_i \quad (6.41)$$

$$\psi'' \equiv \frac{\partial^2\psi}{\partial c_i \partial c_j} = (\hat{M} + \alpha\hat{I})_{ij} \quad (6.42)$$

and obtain a new c as

$$c(\text{new}) = c(\text{old}) - \gamma\Delta, \quad (6.43)$$

where $\Delta = \psi'/\psi''$ and γ is the first of $(\frac{1}{2})^0, (\frac{1}{2})^1, (\frac{1}{2})^2, \dots$, such that

$$\psi(c(\text{new})) < \psi(c(\text{old})). \quad (6.44)$$

The search is typically terminated when

$$\frac{\|(\hat{M} + \alpha \hat{I})c - g\|}{\|g\|} \leq 10^{-6}, \quad (6.45)$$

an arbitrary selected tolerance.

To search for the optimal α , one uses the fact that $\xi(\alpha)$ is a monotonic increasing function and designs a simple scheme to obtain

$$\xi(\alpha_{opt}) = \alpha_{opt}^2(c \cdot c) = n\sigma^2. \quad (6.46)$$

There are many other “norm” smoothing methods. One of such methods is described by Freedman [88].

6.2.3 Curvature smoothing

The smoothing parameter α for *curvature smoothing* is determined empirically by plotting the standard deviation of the fit as a function of α . The optimal α is chosen where the rate of increase of the standard deviation starts to increase dramatically. However, there seems to be no satisfactory way of handling noisy data using the curvature smoothing method. When applied to noisy data, much larger α is often required to get the right porosity, which results in an oversmoothed T_2 distribution.

One of the common procedures in the application of *curvature smoothing* is to use the Levenberg-Marquardt method to minimize Eq. (6.20) [180]. The idea is to “walk” from the point of initial guess for f to the point where the function ϕ of Eq. (6.20) is a minimum. This can be accomplished by assuming ϕ is quite close to the minimum and has a quadratic form as follows:

$$\phi(f + \Delta f) = \phi(f) - \beta \cdot \Delta f + \frac{1}{2} \Delta f^\dagger \cdot \omega \cdot \Delta f \quad (6.47)$$

where

$$\beta_x = -\frac{\partial \phi}{\partial f_x}, \quad \omega_{xy} = \frac{\partial^2 \phi}{\partial f_x \partial f_y}, \quad (6.48)$$

and $f + \Delta f$ represents the final correct solution, f the solution before it reaches the final answer. At the point of correct solution, we have

$$\frac{\partial \phi(f + \Delta f)}{\partial \Delta f} = 0 \quad (6.49)$$

which leads to

$$\omega \cdot \Delta f = \beta. \quad (6.50)$$

From (6.50), we can solve for Δf . This is called the inverse-Hessian method. This would have taken us to the minimum point of ϕ if we were very close to the minimum. But the function is often not exactly quadratic and the initial guess for f can be far from the minimum. The safest way to numerically “*descend*” to a lower value for ϕ is to “*walk*” along the negative gradient of ϕ , i.e.,

$$\Delta f_x = \text{constant} \times \beta_x, \quad (6.51)$$

which is a method known as *steepest descent*. The Levenberg-Marquardt method provides a smooth switching between the extremes of the inverse-Hessian method and the steepest descent method by introducing a λ parameter as follows:

$$\omega'_{xy} = \omega_{xy}(1 + \lambda \delta_{xy}). \quad (6.52)$$

The Δf_x is solved from

$$\omega'_{xy} \Delta f_y = \beta_x, \quad f_x(\text{new}) = f_x(\text{old}) + \Delta f_x. \quad (6.53)$$

We then use the new f_x to evaluate ϕ of (6.20). Initially, the λ is set to an arbitrarily small number. If the new value for ϕ is greater than the previous value for ϕ , we increase λ by a factor of 10 and repeat the process until the new ϕ becomes less than the old ϕ . (By increasing λ , we are limiting the step size of the “*walk*”, converging along the steepest descent, and ensuring a decreasing ϕ .) When this is satisfied, the new f_x is accepted, and the process is repeated to find the new f_x by solving (6.53) with a new λ decreased by a factor of 10. This pushes the solution towards the inverse-Hessian method and speeds up the iteration process. To ensure the non-negativity requirement for f_x during the iterations, any new f_x which is negative is set to zero, before the evaluation of ϕ . The iteration process stops when the fractional change of ϕ is less than a selected tolerance, typically 10^{-5} .

6.2.4 Uniform penalty method

Recently, Borgia, Brown, and Fantazzini [34] devised an elaborate scheme for “curvature smoothing” by allowing the regularization parameter “ α ” to be a variable to provide a uniform penalty (UPEN) along the relaxation time axis. A feedback mechanism is built in such that the weight of the penalty term approaches a uniform value throughout all the relaxation times. The net effect of such a feedback is to make sharp peaks sharper and smooth bumps smoother along the T_2 relaxation time axis.

UPEN uses some global smoothing, or regularizing parameters, which are normally not changed from sample to sample. It uses several computed noise parameters in the regularization process, however. It also allows data windowing, with corresponding weighting factors, for T_2 data sets with large numbers of echoes.

The ability of UPEN to fit relaxation data corresponding to distributions containing both sharp and broad features makes it possible in many cases to identify systematic errors in data that can lead to an inappropriate interpretation. For instance, spurious resolved peaks are sometimes produced by data distortion corresponding to a slight downward bending of the logarithmic decay curve. Non-random errors of fit are likely to result, and in many cases there is a significant *cost* for application of the usual non-negativity constraint, in which case the program posts a warning.

With good relaxation data the non-negativity constraint is usually not needed in UPEN if there are no peaks narrow enough to have only a few computed points and if there are no very sharp valleys. By *good relaxation data* is meant data representing a sum of positive exponential components, with random noise added. Most good petrophysical relaxation data do not require the non-negativity constraint with UPEN.

An awkward feature of UPEN is due to the need to include in the feedback process the nearest neighbors of each point on the computed distribution. This can cause the iterative procedure not to converge. In such cases, it usually wanders around a small region of solution space where changes are not significant for interpretation purposes. A smooth procedure for promoting convergence is implemented, if needed after many iterations, by retaining the smaller of the last two penalty values. This brings rapid convergence without significantly changing the computed distribution. In most cases, the iterations converge without this procedure.

6.2.5 Basis function method

In the simple “norm”, “slope”, and “curvature” smoothing methods, the penalty terms are added to suppress the oscillatory behavior of the unknown function $f(s)$. However, such suppression is not always successful especially for high signal-to-noise data where the high precision of the measurements of $g(t)$ allows smaller eigenvalues to enter the inversion process. We can either use a regularizing process, such as uniform penalty method, with a specific feedback algorithm to suppress the oscillatory behavior of $f(s)$, or use a basis function with a broad base to exclude solutions with rapid oscillatory features.

In solving the Fredholm equation of the first kind in discrete form, such as Eq. (6.17), we have assumed $K_{ij} = e^{-t_i/T_j}$ and solved for the corresponding amplitude f_j , in other words, a single amplitude f_j for a single relaxation time T_j . The determination of f_j does not directly affect the amplitude of neighboring T_j 's, i.e. T_{j-1} and T_{j+1} , etc. This is the reason why for high signal-to-noise data, it becomes difficult to suppress the oscillatory behavior of f . To prevent this from happening, one can couple the determination of f_j to the those of the neighboring T_j s. One simple example is that f_j , instead of being the amplitude for e^{-t/T_j} , becomes the amplitude for

$$\frac{1}{2}e^{-t/T_{j-1}} + e^{-t/T_j} + \frac{1}{2}e^{-t/T_{j+1}} \quad (6.54)$$

Thus the determination of f_j has an effect on the amplitudes of the neighboring relaxation times, and vice versa. Having a basis function to represent each f_j forces the solution to have a smooth behavior.

Eq. (6.54) is just an example for the basis function. Watson et al. [142] use the B -splines as basis functions, where the magnetization $M(t_i)/M_0$ is expressed as:

$$M(t_i)/M_0 = g_i = \sum_{j=1}^m C_j \int_0^\infty B_j^p(T) e^{-t_i/T} dT \quad (6.55)$$

The regularization process is carried out by minimizing the following expression:

$$\min \left\{ \sum_{i=1}^n \left(\sum_{j=1}^m C_j \int B_j^p(T) e^{-t_i/T} dT - g_i \right)^2 + \lambda \sum_{j=1}^m (C_{j+1} + C_{j-1} - 2C_j)^2 \right\} \quad (6.56)$$

where curvature smoothing is used.

In fact, if the basis function is broad enough, the *regularizing* action is already included. A penalty term is not needed. The smoothing of the solution f , is to some degree, determined by the broadness of the basis function. Recently, Miller et al. [162] has shown that using gamma functions as basis functions can lead to a simple analytic form for the linear algebraic equations.

The magnetization $M(t_i)/M_0$ is expressed as :

$$M(t_i)/M_0 = g_i = \sum_{j=1}^m A_j \int F_j(T) e^{-t_i/T} dT = \sum A_j I_j \quad (6.57)$$

where the basis function is given by:

$$F_j(s) = (\kappa s/s_j)^\kappa \exp(-\kappa s/s_j)/(s\Gamma(\kappa)) \quad (6.58)$$

where $s = T^{-1}$, and κ is a constant which determines the broadness of the basis function. As κ increases, the spread of the basis function decreases.

This choice of the basis functions results in a simple analytic form for the integrals, I_j ;

$$I_j(t_i) = \left(1 + \frac{s_j t_i}{\kappa}\right)^{-\kappa} \quad (6.59)$$

which fits in nicely in a set of linear algebraic equations.

6.2.6 Singular value decomposition

One of the popular methods for treating the linear inversion problem is the “*singular value decomposition*” (SVD) method. The SVD method is particularly attractive for the following reasons: (1) it does not require a penalty term for smoothing or regularization, thus obviating the need to search for an optimum regularization parameter, (2) it provides a clear picture of the characteristics of the matrix, as well as the overall properties of the inversion procedure, and (3) it offers a straightforward and easily comprehensible method for obtaining a “regularized” solution by removing the small singular values in a fashion that reflects the measurement error.

In general, we assume that the measurement error is the same for every spin echo. Thus, σ_i in Eq. (6.13) is the same for all i . However, the procedure becomes somewhat unclear when we need to process simultaneously two or more NMR data sets, with very different measurement errors, coming from the same sample. In fact, one encounters a similar problem when the

NMR time domain data are compressed or averaged within different window lengths. This windowing leads to different improved precisions.

In the following, we take a closer look at the procedure used in the SVD inversion as to how the noise level affects the solution. A geometrical interpretation, which provides an intuitive picture for some of the formal mathematical considerations described below, can be found in Ref. [79].

Without loss of generality, we shall limit the following discussion to the simultaneous processing of two data sets with different measurement errors. Suppose we need to minimize the following two sets of data simultaneously:

$$\min\left\{\sum_{i=1}^p \frac{1}{\sigma_1^2} \left(\sum_{j=1}^m k_{ij} y_j - b_i\right)^2\right\} \quad (6.60)$$

$$\min\left\{\sum_{i=1}^q \frac{1}{\sigma_2^2} \left(\sum_{j=1}^m l_{ij} y_j - c_i\right)^2\right\} \quad (6.61)$$

where

$$k_{ij} = (1 - e^{-RD_1/rT_j})e^{-t_i/T_j} \quad (6.62)$$

$$l_{ij} = (1 - e^{-RD_2/rT_j})e^{-t_i/T_j}; \quad (6.63)$$

RD_1 and RD_2 represent the RD (repeat delay time or recovery time) of runs 1 and 2; σ_1 and σ_2 , the corresponding measurement errors; b_i and c_i , the corresponding echo trains; and p and q , the corresponding number of echoes (or data points) in the respective runs. We want to solve both data sets simultaneously so that they yield the same solution y_j . We shall call this procedure the “*composite-data processing*.”

To set up the combined data sets as one least squares fit problem, we write

$$\begin{aligned} & \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1m} \\ \vdots & & & \vdots \\ \vdots & & & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nm} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ \vdots \\ y_m \end{bmatrix} \\ &= \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1m} \\ \vdots & & & \vdots \\ \frac{\sigma_1}{\sigma_2} l_{11} & \frac{\sigma_1}{\sigma_2} l_{12} & \dots & \frac{\sigma_1}{\sigma_2} l_{1m} \\ \vdots & & & \vdots \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ \frac{\sigma_1}{\sigma_2} c_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ \vdots \\ b_n \end{bmatrix} \equiv \mathbf{b}, \quad (6.64) \end{aligned}$$

where A_{ij} are the elements of the combined $m \times n$ matrix $\mathbf{A}$ with $n = p + q$, and $\mathbf{b}$ is the combined data vector with n elements.

Note that the data for run 2, as well as the corresponding matrix elements, have now been scaled by a factor of σ_1/σ_2 , so that the combined data set has a common noise level of σ_1 . The latter will be used to determine the cutoff in the SVD inversion procedure.

Once the combined data sets are set up as one least squares fit problem with the proper weighting taken care of, the method for solving the equation need not to be limited to the SVD technique. The combined data set need not be restricted to two data sets, either.

In fact, we encounter a similar situation of different variances when compressing the time domain data into several windows to increase the computation speed. For example, Eq. (6.13) can be transformed to

$$\sum_{i=1}^s \frac{1}{N_i \sigma^2} \left(\sum_{j=1}^m f_j K_{ij} - B_i \right)^2, \quad (6.65)$$

where n echoes have been partitioned into s windows ($i = 1, \dots, s$) with N_i echoes in the i -th window, and

$$\begin{aligned} n &= N_1 + \dots + N_{i-1} + N_i + \dots + N_s, \\ r_i &= N_1 + \dots + N_{i-1}, \quad r_1 = 0, \\ K_{ij} &= \sum_{k=r_i+1}^{r_i+N_i} e^{-t_k/T_j}, \\ B_i &= \sum_{k=r_i+1}^{r_i+N_i} b_k. \end{aligned} \quad (6.66)$$

Now the variance for B_i is $N_i \sigma^2$, and each window has a different variance.

Similarly, we can also average the data within each window, assuming that arithmetic averaging is a valid procedure for this type of data, to obtain

$$\sum_{i=1}^s \frac{1}{(\sigma/\sqrt{N_i})^2} \left(\sum_{j=1}^m f_j e^{-t_i/T_j} - \bar{b}_i \right)^2, \quad (6.67)$$

where $\bar{b}_i$ is simply the average of the data within the i -th window, and t_i is an appropriate mid-point within that window. The noise level for the i -th window is now reduced to $\sigma/\sqrt{N_i}$.

Regardless of time domain data sum or average, it is easy to rescale the matrix elements and the data vectors so that all windows have the same noise level, as we did in the composite-data processing. This common noise level can then be used to determine the cutoff in the SVD inversion procedure.

Now that a common noise level for a composite-data set is established, we need to determine how to cut off the singular values in the SVD inversion process. The singular value decomposition theorem in linear algebra [131, 140, 180] states that any real $n \times m$ matrix A ($n \geq m$) can be written as the product of an $n \times n$ orthonormal matrix U , an $n \times m$ diagonal matrix Λ with positive or zero diagonal elements, and the transpose of an $m \times m$ orthonormal matrix V , i.e.

$$A = U\Lambda V^T \quad (6.68)$$

where

$$U^T U = I, \quad V^T V = I \quad (6.69)$$

$$\sum_{i=1}^n U_{ip} U_{iq} = \delta_{pq}, \quad 1 \leq p, q \leq n \quad (6.70)$$

$$\sum_{i=1}^m V_{ip} V_{iq} = \delta_{pq}, \quad 1 \leq p, q \leq m \quad (6.71)$$

and

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m). \quad (6.72)$$

The diagonal elements $\lambda_1 > \lambda_2 > \dots > \lambda_m \geq 0$ are called the “singular values” of the matrix A .

The straightforward Householder transformations used for the decomposition of A need not produce all positive singular values λ 's in descending order. However, it is easy to make them all positive by changing the signs of those that are negative and changing the signs of the corresponding eigenvectors, i.e., either the column vectors u_i of U or the column vectors v_i of V . It is also straightforward to rearrange the singular values in a descending order by rearranging the corresponding columns of U and V . Once the program is written in such a way as to produce a Λ with all the singular values positive and in descending order, the information about the ordering of the eigenvectors is already contained in both U and V matrices.

Eq. (6.64) can be written concisely, using vector/matrix notation, as

$$Ay = b \quad (6.73)$$

where the matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$ operates on a vector $\mathbf{y} \in \mathbb{R}^m$ to produce a vector $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{A}$ has all the properties described in Eqs. (6.68-6.72). Then

$$\|\mathbf{A}\mathbf{y}\|^2 = \mathbf{y}^T \mathbf{A}^T \mathbf{A} \mathbf{y} = \mathbf{y}'^T \boldsymbol{\Lambda}^T \boldsymbol{\Lambda} \mathbf{y}' = \|\boldsymbol{\Lambda} \mathbf{y}'\|^2 \quad (6.74)$$

where $\mathbf{A} = \mathbf{U}\mathbf{A}\mathbf{V}^T$, $\mathbf{y} = \mathbf{V}\mathbf{y}'$, and $\|\cdot\|^2$ represents the squared norm of a vector, i.e., the sum of squares of all its components.

The largest (smallest) value of $\|\mathbf{A}\mathbf{y}\|^2/\|\mathbf{y}\|^2$ occurs when $\mathbf{y}$ lies along the direction of the eigenvector of the square orthogonal matrix $\mathbf{V}$ (actually, a column of that matrix) corresponding to the largest (smallest) singular value λ_1 (λ_m). Hence, for an arbitrary vector $\mathbf{y}$, the following inequality is established:

$$\lambda_1^2 \geq \frac{\|\mathbf{A}\mathbf{y}\|^2}{\|\mathbf{y}\|^2} \geq \lambda_m^2, \quad (6.75)$$

which means, for the vector $\mathbf{y}$ which solves Eq. (6.73):

$$\lambda_1^2 \geq \frac{\|\mathbf{b}\|^2}{\|\mathbf{y}\|^2} \geq \lambda_m^2. \quad (6.76)$$

Frequently, we measure $\mathbf{b}$ and try to determine $\mathbf{y}$. There is always measurement error involved in obtaining $\mathbf{b}$. Therefore, instead of $\mathbf{b}$ and $\mathbf{y}$, we have $\mathbf{B} = \mathbf{b} + \delta\mathbf{b}$ and $\mathbf{Y} = \mathbf{y} + \delta\mathbf{y}$ and are faced with the problem of solving

$$\mathbf{A}(\mathbf{y} + \delta\mathbf{y}) = \mathbf{b} + \delta\mathbf{b} \quad (6.77)$$

rather than Eq. (6.73): $\mathbf{A}\mathbf{y} = \mathbf{b}$. We note that $\mathbf{y}$ and $\mathbf{b}$ represent, respectively, the ideal solution and the ideal data vector without measurement errors, while $\delta\mathbf{y}$ and $\delta\mathbf{b}$ represent, respectively, the deviation in the solution and the measurement error of the data vector.

In order to satisfy both Eqs. (6.77) and (6.73), we must have

$$\mathbf{A}\delta\mathbf{y} = \delta\mathbf{b}. \quad (6.78)$$

Assuming that the matrix $\mathbf{A}$ is of rank m , its form in the principal coordinates or principal axes is represented by $\boldsymbol{\Lambda}$, with m non-vanishing descending singular values $\lambda_1 \dots \lambda_m$, and the normalized left and right singular vectors are denoted by $\mathbf{u}_1 \dots \mathbf{u}_n$, and $\mathbf{v}_1 \dots \mathbf{v}_m$, respectively. We can then transform $\mathbf{y}$ and $\mathbf{b}$ to the principal axes as $\mathbf{y}'$ and $\mathbf{b}'$:

$$\mathbf{y}' = \mathbf{V}^T \mathbf{y} \quad (6.79)$$

$$\mathbf{b}' = \mathbf{U}^T \mathbf{b}, \quad (6.80)$$

where the norms or lengths of both $\mathbf{y}$ and $\mathbf{b}$ are left unchanged by these orthogonal transformations, i.e., $\|\mathbf{y}\| = \|\mathbf{y}'\|$ and $\|\mathbf{b}\| = \|\mathbf{b}'\|$.

Note that since the operation of $\mathbf{A}$ is restricted to the space spanned by the eigenvectors $\mathbf{v}_1 \dots \mathbf{v}_m$ of $\mathbf{V}$, both $\mathbf{b}$ and $\mathbf{b}'$ must lie in the subspace spanned by an appropriate m dimensional subspace of the space spanned by the totality of eigenvectors $\mathbf{u}_1 \dots \mathbf{u}_n$ of $\mathbf{U}$, if the model for inversion is correct. Any portion of either $\mathbf{b}$ or $\mathbf{b}'$, which lies in the complementary $n - m$ dimensional orthogonal subspace of the eigenvectors of $\mathbf{U}$, is due to inaccuracies of this model.

In the principal coordinates we have

$$\Lambda \mathbf{y}' = \Lambda \mathbf{V}^T \mathbf{y} = \mathbf{U}^T \mathbf{b} = \mathbf{b}' \quad (6.81)$$

$$\Lambda \delta \mathbf{y}' = \Lambda \mathbf{V}^T \delta \mathbf{y} = \mathbf{U}^T \delta \mathbf{b} = \delta \mathbf{b}'. \quad (6.82)$$

Assuming that different components b_i of the data vector $\mathbf{b}$ have errors that are statistically uncorrelated and have the same variance σ^2 , namely

$$\langle \delta b_i \delta b_j \rangle = \sigma^2 \delta_{ij}, \quad (6.83)$$

we can write

$$\langle \delta b'_i \delta b'_j \rangle = \left\langle \sum_k U_{ki} \delta b_k \sum_l U_{lj} \delta b_l \right\rangle = \sigma^2 \delta_{ij} \quad (6.84)$$

$$\langle \delta y'_i \delta y'_j \rangle = \left\langle \frac{\delta b'_i}{\lambda_i} \frac{\delta b'_j}{\lambda_j} \right\rangle = \frac{\sigma^2}{\lambda_i^2} \delta_{ij} \quad (6.85)$$

$$\langle \delta y_i \delta y_j \rangle = \left\langle \sum_k V_{ik} \delta y'_k \sum_l V_{jl} \delta y'_l \right\rangle = \sum_k \frac{\sigma^2}{\lambda_k^2} V_{ik} V_{jk}. \quad (6.86)$$

The requirement for a good solution, $\langle \delta y_i^2 \rangle \ll y_i^2$, is thus translated into the following inequality:

$$\langle \delta y_i^2 \rangle = \sum_k \frac{\sigma^2}{\lambda_k^2} V_{ik}^2 \ll y_i^2 = \left(\sum_k V_{ik} y'_k \right)^2 = \left(\sum_k \frac{V_{ik}}{\lambda_k} b'_k \right)^2 = \left(\sum_{kl} \frac{V_{ik} U_{kl}}{\lambda_k} b_l \right)^2. \quad (6.87)$$

A necessary condition for this to be true for every i is that the sums over i also obey the same inequality

$$\sum_{i=1}^m \langle \delta y_i^2 \rangle \ll \sum_{i=1}^m y_i^2. \quad (6.88)$$

The r.h.s of the last inequality is

$$\sum_{i=1}^m y_i^2 = \sum_{i=1}^m y_i'^2 = \sum_{i=1}^m \frac{b_i'^2}{\lambda_i^2}, \quad (6.89)$$

and its l.h.s is

$$\sum_{i=1}^m \langle \delta y_i^2 \rangle = \sum_{i=1}^m \langle \delta y_i'^2 \rangle = \sum_{i=1}^m \frac{\langle \delta b_i'^2 \rangle}{\lambda_i^2} = \sum_{i=1}^m \frac{\sigma^2}{\lambda_i^2}. \quad (6.90)$$

Thus we need to satisfy

$$\sum_{i=1}^m \frac{\sigma^2}{\lambda_i^2} \ll \sum_{i=1}^m \frac{b_i'^2}{\lambda_i^2}. \quad (6.91)$$

This inequality will be satisfied if $\sigma^2 \ll b_i'^2$ for all i . However, it can fail if the m supposedly independent equations are actually nearly non-independent, which results in some very small singular values λ_i (see Ref. [180]), and if the corresponding b_i' are not much greater than the error estimate σ . Those terms will then dominate the two sums of Eq. (6.91), and the inequality will not be satisfied. This failure can be remedied by discarding a sufficient number of terms with small λ_i , e.g., all λ_i , $i > r$, such that the more restricted sums (i.e., for $i = 1 \dots r$; $r \leq m$) satisfy the inequality to the desired degree.

A practical way to choose a suitable lower cutoff λ_r for the singular values is obtained from the following considerations: If the sums in Eq. (6.91) are restricted to run from 0 up to $r \leq m$, and if $\lambda_1 \gg \lambda_r$, then the two sides of that inequality should satisfy the following inequalities:

$$\sum_{i=1}^r \frac{\sigma^2}{\lambda_i^2} \ll \frac{\sigma^2}{\lambda_r^2}, \quad \sum_{i=1}^r \frac{b_i'^2}{\lambda_i^2} \gg \frac{\sum_{i=1}^r b_i'^2}{\lambda_1^2} \approx \frac{\sum_{i=1}^m b_i'^2}{\lambda_1^2} = \frac{\|\mathbf{b}'\|^2}{\lambda_1^2} = \frac{\|\mathbf{b}\|^2}{\lambda_1^2}. \quad (6.92)$$

Here we assumed that the discarded components $b_{r+1}' \dots b_m'$ make a negligible contribution to the norm $\|\mathbf{b}'\| = \|\mathbf{b}\|$. Presumably, this would follow if those components are of order σ , which should be much less than the norm itself. If we choose r by demanding that the right-hand-sides of these two inequalities be equal, namely

$$\frac{\lambda_1}{\lambda_r} = \frac{\|\mathbf{b}'\|}{\sigma} = \frac{\|\mathbf{b}\|}{\sigma} \approx \frac{\|\mathbf{b} + \delta\mathbf{b}\|}{\sigma}, \quad (6.93)$$

then the inequality

$$\sum_{i=1}^r \frac{\sigma^2}{\lambda_i^2} \ll \sum_{i=1}^r \frac{b_i'^2}{\lambda_i^2}, \quad (6.94)$$

which replaces Eq. (6.91), is ensured. Once a proper cutoff such as Eq. (6.93) is determined, both the variances and the covariances of the uncertainties δy_i can be calculated using Eq. (6.86).

The singular values smaller than λ_r , (i.e., all λ_i , $i > r$) which are discarded in the restricted sums, are effectively replaced by 0 in the inverse matrix Λ^{-1} . This means that the matrix Λ^{-1} is replaced by a matrix of lower rank r , Λ_r^{-1} , which has only r positive (non-zero) diagonal elements, and both U and V are effectively replaced by smaller matrices U_r and V_r , which have the smaller sets of eigenvectors $u_1 \dots u_r$ and $v_1 \dots v_r$ ($r \leq m$), respectively.

Hence the solution $Y = y + \delta y$ is given by

$$Y = A^{-1}B = V_r \Lambda_r^{-1} U_r^T (b + \delta b) \quad (6.95)$$

where $B = b + \delta b$, and where A^{-1} is a pseudo inverse operating only on the subspace spanned by $u_1 \dots u_r$ and leading to results in the subspace spanned by $v_1 \dots v_r$. The measurement error σ is taken into consideration by discarding all the singular values that are smaller than λ_r . This means that both AY and b are in the r dimensional subspace spanned by $u_1 \dots u_r$, while δb is in the complementary $m - r$ dimensional orthogonal subspace. Similarly, Y is in the r dimensional subspace spanned by $v_1 \dots v_r$, while δy is in the complementary $m - r$ dimensional orthogonal subspace. Therefore, we have

$$\|b + \delta b\|^2 = \|b\|^2 + \|\delta b\|^2 \quad (6.96)$$

and

$$\|AY - B\|^2 = \|AY - b\|^2 + \|\delta b\|^2, \quad (6.97)$$

and the l.h.s. is at its minimum when $AY - b = 0$. When there is no constraint on Y , a solution for $AY - b = 0$ can always be found. However, if some of the elements of Y happen to be negative, and the non-negativity constraint is imposed, then Y is no longer at the calculus minimum of $\|AY - b\|^2$ and $\|AY - b\|^2 \geq 0$ at the minimum.